



**EXPLORATORY STUDY TO DETERMINE EFFICACY OF
HAPTIC FEEDBACK AND VIRTUAL REALITY FOR FLIGHT
SIMULATION TRAINING**

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**EXPLORATORY STUDY TO DETERMINE EFFICACY OF HAPTIC
FEEDBACK AND VIRTUAL REALITY FOR FLIGHT SIMULATION
TRAINING**

The Capstone Design Project
presented to the academic faculty

by

Student A

Student B

in partial fulfillment of the requirements for
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Habib University

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FEEDBACK AND VIRTUAL REALITY FOR FLIGHT SIMULATION
TRAINING**

This capstone design project was advised by:

Dr. BlaBla
Faculty of Electrical Engineering
Habib University

Approved by the Faculty of Electrical Engineering on April 19, 2019.

A great quote to start the thesis

Einstein

A great dedication goes here.

ACKNOWLEDGEMENTS

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EXECUTIVE SUMMARY

This summary should be a stand-alone part of your document. The reader should be able to comprehend this even if it were circulated separately from the rest of document. This should ideally be one page long only and could be read in place of the complete report. You should highlight the problem you're trying to solve, major technical challenges involved, your approach or specific solution, key features of your solution, your proof of concept, and the way forward.

KEYWORDS: Awesomeness, Technical Marvel

CHAPTER 1

INTRODUCTION AND BACKGROUND

1.1 Introduction

What is the problem in the broader context? Provide a description of the problem in simple language, explaining problem to a layman. What is the motivation or need to solve this problem? Provide context or background information. Discuss the desired functions and features in a possible solution. Discuss the technical challenges slowing development of solution. In short, bring the reader to the playground on which you intend to play.

1.2 Literature Review

Provide a review of related solutions to the problem.

1.3 Problem Statement/Objective

State now in a clear and concise manner the objective of your capstone project, in particular. This could be a very small part of the broader problem you have identified in the previous section, e.g. you could be tackling only one particular technical challenge of the many identified.

CHAPTER 2

DESIGN PROCESS

This chapter should include the following:

2.1 Clients/Stakeholders and their requirements

Discuss the stakeholders. Provide a list of requirements (features) you developed from the viewpoint of the client(s)/stakeholders. Describe the objectives considered for your project and their corresponding metrics, e.g. minimizing cost could be an objective. Discuss the functions that your product/solution should perform. These would typically be phrased as: our solution/product will be able to .. Describe any design constraints imposed by the stakeholders. Discuss any interviews or other research conducted with client(s)/stakeholders in this section. Include the research results either in the main body or as an appendix.

2.2 Design Concepts

Discuss the different concept ideas generated during the ideation phase. You can describe the different concepts for fulfilling each function, followed by discussion of the different integrated solutions possible as a result of these choices. Discuss how each solution performs in terms of your design specifications and objectives. Present the results of your comparison. Feel free to use an appendix to include additional details.

2.3 Society, Economic, and Ethical considerations

Discuss the impacts for your designs in the areas of society, health, economics, and manufacturability. Provide any relevant engineering or other standards which are applicable for your project. Discuss how these factors (if any) played a role in choosing the final design.

2.4 Environment and Sustainability considerations

Discuss the impacts for your design in the areas of environment and sustainability. Also, discuss how your design evolved in light of these considerations.

2.5 Technical Requirements

Provide an overview of your final design (save details for later) and how it compares to existing solutions. Finally provide quantitative specifications for the features in your solution and the means through which you'll verify, e.g. if your solution will be able to measure the speed of an incoming vehicle then provide the maximum speed it will be able to measure. Discuss the allowable tolerances of your solution, where applicable e.g. the solution will be able to measure speed with at least 80

2.5.1 Example Subsection

This is a subsection in Chapter 2.

Example Subsubsection

This is a subsubsection in Chapter 2.

CHAPTER 3

DESIGN DETAILS

You are required to include the following details in this category.

3.1 Solution Statement

Describe what solution have you worked upon in this thesis Why the proposed solution is best among the existing/alternate solutions Discuss your solution's sustainability and its impact on environment and society

3.2 Solution Overview

Give a brief overview of your solution/system with the help of a Block Diagram Briefly discuss each sub-system

3.3 System Details

Give details of each sub system shown in the block diagram (Each sub-system should be described in details as a separate chapter)

Appendices You may include additional information related to design in Appendix. For example, circuit schematics, PCB layout, BoM (Bill of Materials), software scripts or hardware drivers, etc.

Table 3.1: This is an example Table.

x	f(x)	g(x)
1	6	4
2	6	3
3	6	2
4	6	2

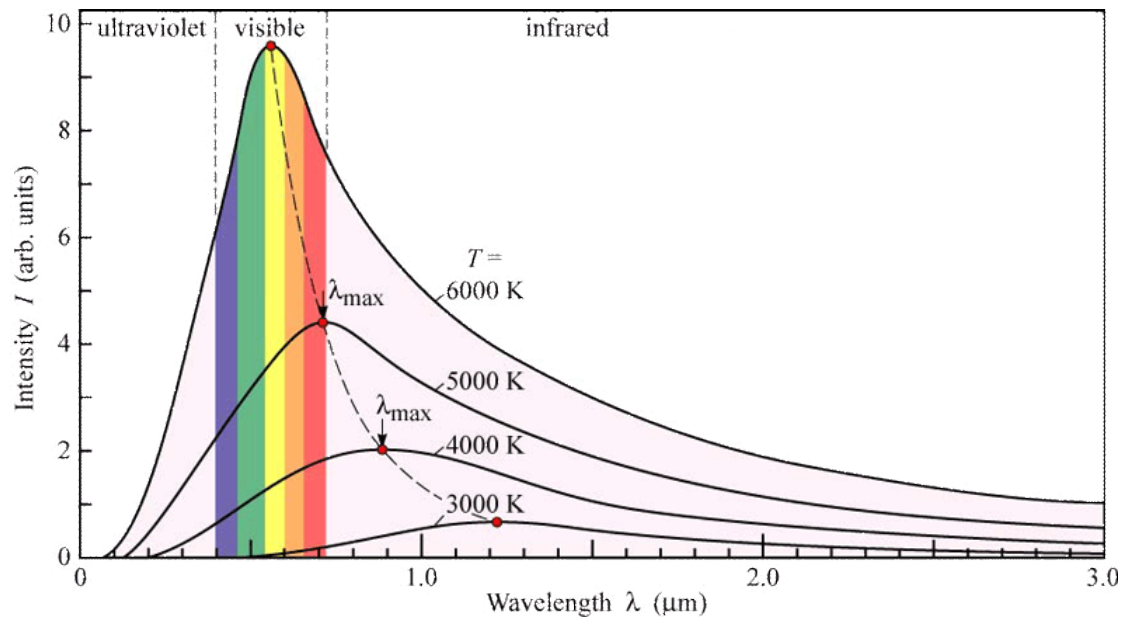


Figure 3.1: This is an example Figure.

This is how you cite [1].

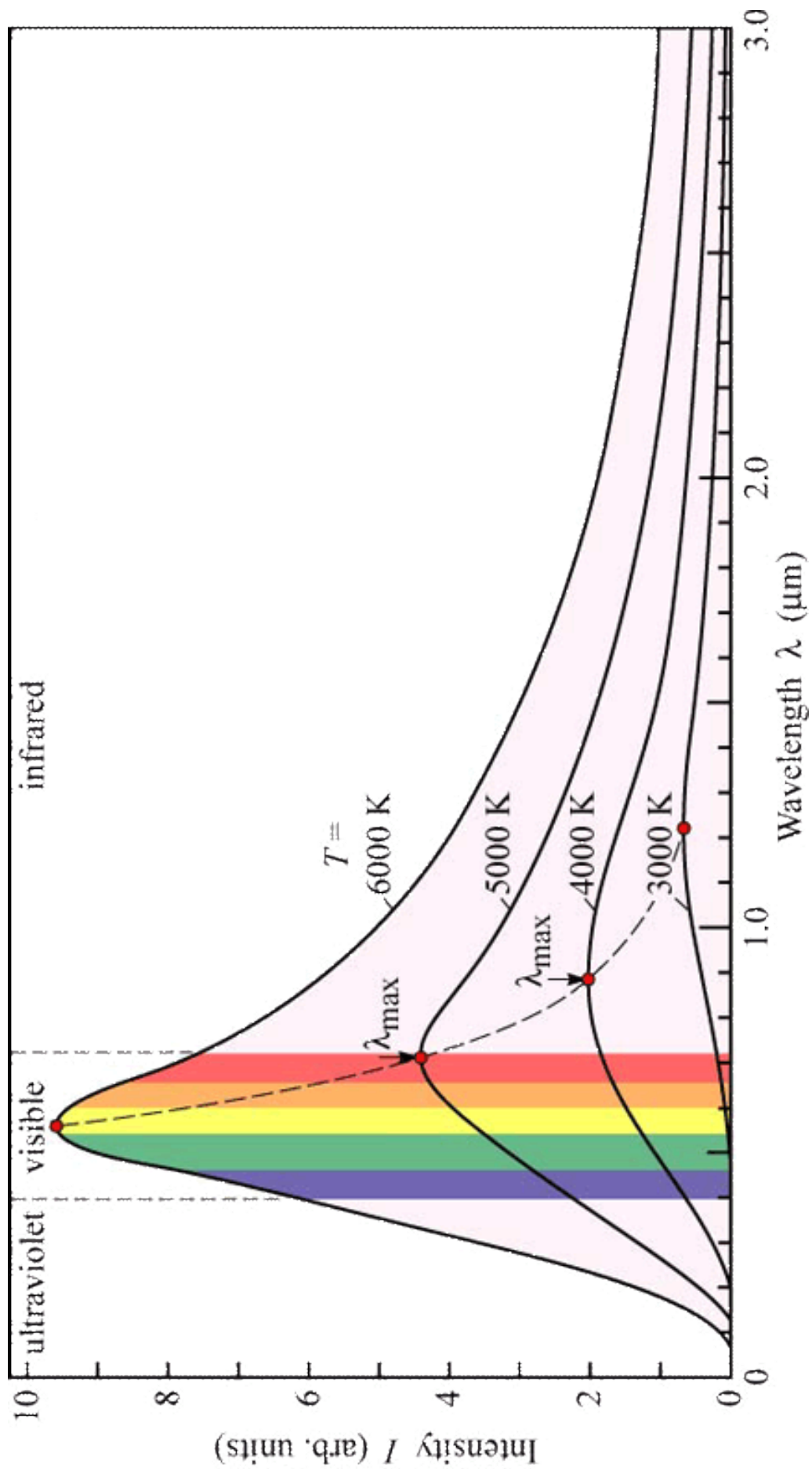


Figure 3.2: This is another example Figure, rotated to landscape orientation.

CHAPTER 4

PROTOTYPING AND TESTING

Discuss how you prototyped your design and evaluated it, i.e. what were your testing procedures and patterns. Also, discuss how will this prototype be scaled to the solution you have outlined in the previous section. This chapter **MUST** contain:

1. Pictures of your prototype
2. Explain in detail your prototype testing procedures/methodologies. The designed testing scenarios **MUST** capture all functionalities of prototype.
3. Explain what Key performance indicators (KPIs) you are considering to evaluate the performance of your prototype
4. Show results based on actual data collected from your prototype. Results should be in form of tables & graphs
5. Compare actual prototype results with theoretical design values. Elaborate this comparative analysis in details & discuss the finding (output matching design values or not, if not then calculate percentage deviation, technical justification of deviation found).
6. In case of deviation in result w.r.t. to design values, if you think there is potential to extend this project in future with possible solution/way forward, then add this part in next chapter “Future & conclusion”

Annexure related to this chapter:

- Include operational details or a manual of your prototype as an appendix. Most of your prototypes will be disassembled for parts, so this manual should include sufficient details to recreate prototype

- Include all drawing /CAD design
- Include code (if necessary)
- HW component details , include all details like data sheet, vendor name & cost.
- Any other data related to your project which can be useful for future use.

CHAPTER 5

CONCLUSION AND FUTURE WORK

Provide a summary of what has been accomplished and your conclusions. Discuss the remaining work (if any) and/or provide suggestions for future work.

Appendices

- Include operational details or a manual of your prototype as an appendix. Most of your prototypes will be disassembled for parts, so this manual should include sufficient details to recreate prototype
- Include all drawings: Circuit schematics, PCB layouts, CAD designs
- Include code (if necessary), software scripts or hardware drivers, etc.
- HW component details , include all details like data sheet, vendor name & cost.
- Any other data related to your project which can be useful for future use.

APPENDIX A
MANUAL OF PROTOTYPE

Include the above distributed in various chapters.

REFERENCES

- [1] B. W. Kernighan and D. M. Ritchie, *The C Programming Language Second Edition*. Prentice-Hall, Inc., 1988.

VITA

The length of the vita is preferably one page. It may include the place of birth and should be written in third person. This vita is similar to the author biography found on book jackets.

